NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Engineering

Department of Civil & Water Engineering, Department of Chemical Engineering, and Department of Industrial Engineering

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COURSE OUTLINE

SCS 1101 – Introduction to Computer Science and Programming

Field	Details
Year	I
Semester	I
Lecturer	Mr. B. Guvava
Office	FD60
Contact Phone	0719 333 660
Email	brian.guvava@nust.ac.zw
Venue	Computer Lab
Credits	10
Contact Hours	0800 – 1600 (Extended 5-Week Intensive)
Personal Hours	52 hours
Prerequisites	None
Co-requisites	None

Synopsis

This module covers the fundamentals of computer science and programming. Students will learn about the Information Society and the History of Computers; Data and Information; Number Systems and Arithmetic; Data Representation; Basic Computer Components (CPU, I/O units, Storage); Brief Concepts of Computer Languages and Programming Techniques including high/low-level languages, compilers, interpreters, grammar, recursion, and simple data structures (arrays, lists, trees, hash tables, queues and stacks); Problem Solving; Algorithms (Sorting, Compression, Numerical and Encryption); Operating Systems and their functions (process and memory management, I/O, Data Communication, Job Control); File Structures, Organization and Access; Databases; Fundamentals of Networks; and a Simple Program including initialization, printing, comments, and keywords.

Description of the Course

This course introduces first-year Civil and Water Engineering students to the foundational principles of computer science and structured programming. Emphasis is placed on understanding how computer systems represent, store, and process data; the

relationships between hardware and software components; and the practical role of computers in engineering and society. Students develop problem-solving skills by studying algorithms and are guided through the design and implementation of simple programs in the C programming language. The course is structured as a five-week intensive, combining daily lectures, laboratory sessions, group research, assignments, and a formal test.

Learning Objectives

At the end of this course, students should be able to:

i. Identify and describe computer hardware components (CPU, I/O units, storage) and explain how data is represented and processed within a computer system.

ii. Distinguish between different levels of programming languages and explain the role of compilers, interpreters, and operating systems in program execution.

iii. Apply algorithmic thinking and problem-solving techniques, including sorting, searching, and recursive algorithms, to design solutions for given problems.

iv. Design, write, and test simple computer programs in C that demonstrate correct use of variables, data structures, control flow, input/output, and comments.

v. Explain the fundamentals of computer networks, database systems, file organization, and data communication, and relate these concepts to real-world engineering applications.

Course Content

TOPIC	WEEK	CONTACT HOURS
1. FOUNDATIONS OF COMPUTER SCIENCE & SOCIETY	**1**	
1.1 Course introduction, materials, and assessment overview		0.5
1.2 The Information Society: role of computers in modern life		1
1.3 History of Computers: generations and evolution		1
1.4 Data vs. Information: concepts and distinctions		1
2. NUMBER SYSTEMS & DATA REPRESENTATION	**1**	
2.1 Number systems: Binary, Octal, Decimal, Hexadecimal		1
2.2 Arithmetic operations in different number bases		1
2.3 Data representation: integers, floating-point, characters (ASCII/Unicode)		1
3. BASIC COMPUTER COMPONENTS	**1**	
3.1 Central Processing Unit (CPU): ALU, Control Unit, Registers		1
3.2 Input and Output units: devices and interfaces		1

TOPIC	WEEK	CONTACT HOURS
3.3 Storage: primary memory (RAM/ROM) and secondary storage		1
3.4 Evaluating and selecting computer hardware		0.5
4. OPERATING SYSTEMS & SYSTEM SOFTWARE	**2**	
4.1 Operating systems overview: roles and types		1
4.2 Process management and memory management		1
4.3 I/O Management and Job Control		1
4.4 File structures: organization, access methods (sequential, random, indexed)		1
5. DATA COMMUNICATION & FUNDAMENTALS OF NETWORKS	**2**	
5.1 Data communication concepts: signals, bandwidth, protocols		1
5.2 Network types: LAN, WAN, MAN; wired and wireless		1
5.3 Internet services and basic network security		1
6. DATABASE FUNDAMENTALS	**2**	
6.1 Introduction to databases and data models		1
6.2 Database Management Systems (DBMS): examples and features		1
6.3 File vs. database systems: advantages and applications		1
7. COMPUTER LANGUAGES & PROGRAMMING CONCEPTS	**3**	
7.1 Low-level languages: machine code and assembly language		1
7.2 High-level languages: overview and classification		1
7.3 Compilers vs. interpreters: roles and differences		1
7.4 Program structure: grammar, syntax, and keywords		1
8. SIMPLE DATA STRUCTURES	**3**	
8.1 Arrays and lists: declaration, access, and use cases		1
8.2 Stacks and queues: LIFO/FIFO principles and applications		1
8.3 Trees and hash tables: introduction and concepts		1
8.4 Recursion: concept, base case, recursive case, examples		1
9. ALGORITHMS & PROBLEM SOLVING	**4**	
9.1 Problem-solving methodology: decomposition and flowcharts		1
9.2 Sorting algorithms: Bubble Sort, Selection Sort, Insertion Sort		2
9.3 Searching algorithms: Linear Search, Binary Search		1
9.4 Compression and encryption: concepts and real-world		1

TOPIC	WEEK	CONTACT HOURS
relevance		
9.5 Numerical algorithms: overview and examples		1
10. INTRODUCTION TO PROGRAMMING PRACTICE	**4**	
10.1 Writing a simple program: structure and workflow		1
10.2 Variables, initialization, data types, and assignment		1
10.3 Input/Output: printing to screen, reading user input		1
10.4 Comments, indentation, and coding standards		0.5
10.5 Control flow: conditionals (if/else) and loops (for/while)		1
11. CONSOLIDATION, ASSESSMENT & GROUP RESEARCH PRESENTATIONS	**5**	
11.1 Group Research Presentations (student-led sessions)		4
11.2 Test 1 (in-class written test)		2
11.3 Assignment submission and review session		1
11.4 Course review, Q&A, and examination preparation		2
TOTAL NUMBER OF CONTACT HOURS		**48**

Forms and Dates of Assessment

CONTINUOUS ASSESSMENT 40%: 1 Test, 2 Assignments, 1 Group Research and Presentation (Dates To Be Advised)

FINAL EXAMINATION 60%

DESCRIPTION	DATES	WEIGHTING	WEEK	HOURS
Personal Study	—	—	1–5	30
Group Research	To Be Advised	10%	1–4	12
Assignment 1	To Be Advised	10%	1–3	3
Assignment 2	To Be Advised	10%	2–4	2
Test 1	To Be Advised	10%	5	2
Examination	—	60%	—	3
TOTAL NUMBER OF COURSE HOURS				**100**

Notes

- No student will be allowed into the Examination Hall without Coursework.
- No plagiarism in assignments; violations will result in disciplinary action.
- Students should attend at least 80% of all lectures and lab sessions.
- Students with special requirements should consult with the lecturer for special arrangements.

Recommended Sources

Textbooks

1. Brookshear, J.G. & Brylow, D. (2019). *Computer Science: An Overview* (13th ed.). Pearson. [Comprehensive introduction to CS fundamentals]
2. Kernighan, B.W. & Ritchie, D.M. (1988). *The C Programming Language* (2nd ed.). Prentice Hall. [Classic programming reference]
3. Hanly, J.R. & Koffman, E.B. (2012). *Problem Solving and Program Design in C* (7th ed.). Addison-Wesley. [Problem-solving and C programming]
4. Forouzan, B.A. & Fegan, S.C. (2012). *Foundations of Computer Science* (3rd ed.). Cengage Learning. [Data representation, algorithms, networking]
5. Laudon, K.C. & Laudon, J.P. (2010). *Management Information Systems: Managing the Digital Firm* (9th ed.). Prentice Hall. [Information systems context]

Internet / Open Resources

Resource	URL / Description
NUST E-Learning (imfundo)	https://imfundo.nust.ac.zw – Course materials, announcements, and submissions
NUStone Digital Library	http://library.nust.ac.zw – Access e-books, journals, and past exam papers
CS50 – Harvard OpenCourseWare	https://cs50.harvard.edu – Free, world-class intro to computer science (video lectures, problem sets)
Khan Academy – Computing	https://www.khanacademy.org/computing – Interactive lessons on algorithms, data structures, and programming
W3Schools	https://www.w3schools.com – Quick reference and tutorials for C, HTML, databases, and networking
C Reference (cppreference)	https://en.cppreference.com – Official C language reference and standard library documentation
TutorialsPoint – C	https://www.tutorialspoint.com/cprogramming – Comprehensive C programming tutorials with examples
Open Directory of Open Access Repositories	http://www.opendoar.org – Find open-access academic papers and textbooks
Visualgo.net	https://visualgo.net – Interactive visualizations of sorting, searching, and data structure algorithms
GeeksforGeeks	https://www.geeksforgeeks.org – Articles and examples on data structures, algorithms, OS, and networking

Consultations

The lecturer is available for consultation on any weekday between 12:00 hrs and 15:30 hrs. Tuesdays and Thursdays are generally preferred. Students are encouraged to arrange appointments via email at brian.guvava@nust.ac.zw.